The form of dissipation shapes what a quantum reservoir remembers

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Dissipation sets how long a quantum reservoir computer remembers its past inputs, and its rate is the usual control. Here we show that its form matters as much, for how a memory is stored, written and read. When the qubits decay independently, stronger loss shortens the memory. When they decay collectively, the quantum Zeno effect holds the system in dark states that do not decay, and stronger loss prolongs the memory. A longer memory is harder to write, because it leaves more time for new inputs to mix with the inputs already stored. We prove that the number of inputs mixed into a stored signal is, on average, the memory lifetime times the fraction of the signal that one input mixes. We also prove that this number stays fixed, however long the memory lasts, when the input amplitude falls as the inverse square root of the lifetime. A weaker input leaves a fainter signal to read out, so the form of the dissipation, the input amplitude and the measurement budget have to be chosen together.

1 Introduction

Processing information over time requires a memory of what came before. A physical system can provide such a memory because each input changes its state and leaves traces that influence its later response. Reservoir computing uses these traces to perform a task: an input stream drives a dynamical system, and a weighted sum of quantities describing its state produces the output. Only these weights are trained; this mapping is called a linear readout [1, 2]. In quantum reservoir computing (QRC), the evolving system is quantum mechanical, and the readout uses quantities obtained by measuring it [3]. The challenge is to keep earlier inputs accessible while new information continues to arrive.

The environment shapes how these traces fade. A reservoir must forget its initial state and old inputs, a property known as fading memory or the echo state property [1, 2], and in a quantum reservoir dissipation provides it [4]. A system can lose energy or information to its surroundings through processes collectively described as dissipation [4–15]. Previous work has shown that tuning the dissipation rate improves memory, in some reservoirs with the best performance at an intermediate loss rate [6, 9]. Input, interactions, and dissipation can also be optimized together [16, 17]. The rate specifies how strongly the environment acts, but the way it acts is a separate design choice. Engineering this interaction can prepare and protect selected quantum states [18–21]. For a system processing an ongoing input stream, this raises the possibility of controlling which traces of the past survive.

This distinction becomes concrete for a reservoir of interacting spins, each a two-level quantum system or qubit. When the spins decay through separate routes into the environment, they undergo independent decay, also called local decay. When they share a common route, they undergo collective decay. In certain joint states, the different emission paths interfere and cancel, preventing decay through that shared route. These protected states are called dark states [22–24]. Internal dynamics can still move the system into states that do decay. If this decay is sufficiently rapid, however, it suppresses those transitions, a phenomenon known as the dissipative quantum Zeno effect [25–28]. Stronger loss therefore slows the escape of information from the protected states (Fig. 1c).

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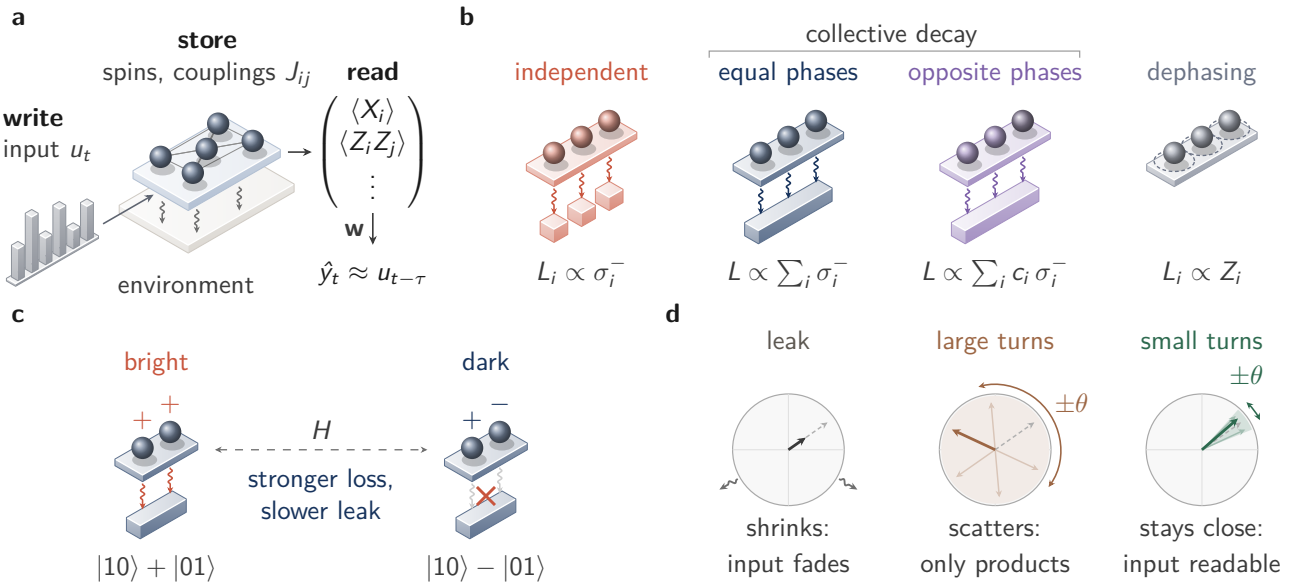


Figure 1: **A quantum reservoir as a memory that is written, stored and read. a** Inputs are written into interacting spins, stored in their state and read by a trained linear readout. **b** Four forms of dissipation with the same total coupling, with the rates of Table 2. The darker spins couple with the opposite sign. **c** Two spins that share one jump operator: their decays add (bright) or cancel (dark), and strong loss holds them in the dark state, the quantum Zeno effect. **d** A stored signal fades because it leaks away or because later inputs turn it at random.

A long-lived state is not yet a usable memory, because each new input changes the same state that stores the past. A signal that initially depends on one input comes to depend on combinations of several inputs, such as their product. This mixing of inputs is the nonlinearity of the reservoir. It is useful for tasks that require relationships between inputs, but it reduces the capacity available for recalling single inputs [29–31]. Reading the memory has a cost as well, because resolving a weak quantum signal requires repeated preparation and measurement [10, 13, 32], and a long-lived memory is hard to read if its signal is faint. Previous work has established memory limits in classical reservoirs [2, 33, 34], explored input strength [35], and investigated how noise and protected states affect quantum information processing [36–39]. Closest to our work, Ref. [6] tunes the dissipation rate of the reservoir we study, Ref. [31] finds one trade-off between memory and nonlinearity across reservoir schemes, and Ref. [39] reads the output of a reservoir from a decoherence-free subspace to classify quantum states. The Zeno protection of dark states and the benefit of a weak input for linear memory are established. How protection, continued input, and measurement together determine the part of the past that a readout recovers is open.

Here we show that the form of the dissipation matters as much as its rate, because it decides how a memory is stored, written and read (Fig. 1). Our contributions are threefold.

(1) *Storing.* Stronger loss can prolong a memory. At equal total coupling, independent decay shortens the memory as the loss grows, whereas collective decay prolongs it, because the quantum Zeno effect holds the stored signal in dark states (Sec. 3.1).

(2) *Writing.* Two theorems say what a long memory holds. The mixing law states that a stored signal combines on average $s\,t_{\mathrm{mem}}$ inputs, the memory lifetime t_{mem} times the fraction s of the signal that one input mixes. The scaling limit states that an input amplitude that falls as $1/\sqrt{t_{\mathrm{mem}}}$ keeps this number fixed however long the memory lasts. The limit agrees with five-spin reservoirs without a fitted parameter, and this amplitude raises their linear memory capacity from 15 or 16 to 40, out of at most 45 (Sec. 3.2).

(3) *Reading.* The stationary state sets what measurements recover. At strong loss a collective channel whose phases differ from those of the input field recovers the most memory, and at weak loss independent decay does (Sec. 3.3).

Both theorems hold for every reservoir that meets

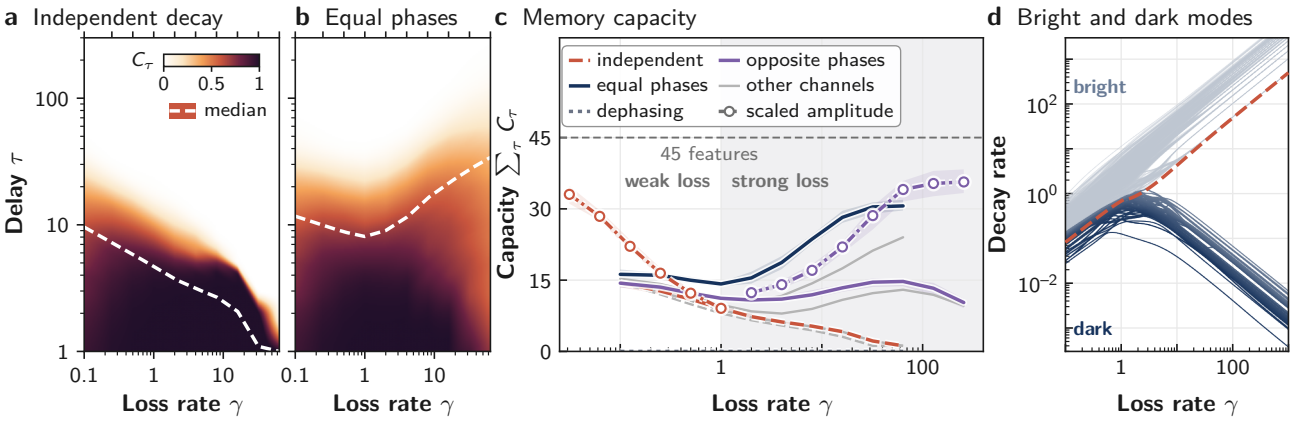


Figure 2: **The decay channel decides whether stronger loss shortens or prolongs the memory. a,b** Recall of the input τ steps back: stronger loss pushes it to shorter delays with independent decay (a) and to longer delays with the equal-phase channel (b). Half of the recall lies below the dashed line. **c** Linear memory capacity of all channels at a fixed input amplitude, and with the scaled amplitude of Sec. 3.2 (circles), in simulated runs summed over the first 300 delays. Shading marks loss stronger than the couplings ($\gamma > 1$). **d** With the equal-phase channel the dark modes decay more slowly as the loss grows, while the slowest mode of independent decay (dashed) decays faster.

their conditions, classical ones included, and give the trade-off between memory and nonlinearity from the equations of motion alone. Together, these results make dissipation a design tool. A reservoir can be built to remember longer under stronger loss, and what it will remember can be computed before it is built.

2 Model

The reservoir is the driven spin system of Ref. [6]. Five spins ($N = 5$) interact through random couplings J_{ij}, and in each step a random input u between 0 and 1 sets the strength of a transverse field for a time Δt:

$$H(u) = \sum_{i<j} J_{ij} X_i X_j + h \sum_i Z_i + h(1+u) \sum_i X_i, \tag{1}$$

with Pauli operators X_i, Z_i of spin i and a field strength h. To weaken the input at a fixed mean drive, we replace $h(1+u)$ by $h[\frac{3}{2} + \eta(u - \frac{1}{2})]$. The amplitude η, known as the input scaling in reservoir computing [35], sets how far each input moves the field. The state ρ of the spins follows a Lindblad master equation [40, 41],

$$\dot{\rho} = -i[H(u), \rho] + \sum_k \Big(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \Big), \tag{2}$$

where the Hamiltonian drives the internal dynamics and each jump operator L_k, scaled by one loss rate γ, is one way the environment changes the state, such as removing an excitation or erasing a phase.

A linear readout is trained on 45 features, the expectation values of X_i, Y_i, Z_i and of the same-axis pairs X_iX_j, Y_iY_j, Z_iZ_j, to estimate the input from τ steps back. Its accuracy is the memory function [33] $C_\tau = \mathrm{corr}^2(\hat{y}_t, u_{t-\tau})$, the squared correlation between the estimate $\hat{y}_t$ and the input on held-out data. We call C_τ the recall at delay τ. Its sum over the delays is the linear memory capacity, and one unit is one past input recalled perfectly. The capacities for all functions of the past inputs add up to the number of features, the total capacity of 45 units [29].

We call the part of the state that depends on the past inputs the stored signal. The variance that an input leaves in the state, taken over the inputs that follow, decays in the long run as $(1-1/t_{\mathrm{mem}})^t$, which defines the memory lifetime t_{mem} in steps (Appendix A.4). It depends on the dissipation and on the inputs, and both theorems use it. Where only the dissipation matters we use the relaxation time t_{rel}, the decay time of the slow modes at the mean input, which for independent decay is the inverse of the Liouvillian gap (Table 1).

Conventions. Capacities quoted in the text come from exact moments, that is, from exact expectation values, the limit of infinitely many shots, summed over all delays. Figure 2c, the tables and the appendix figures show simulated runs summed over the first 300 delays. Section 3.3 counts shots. All numbers are for five spins with

all-to-all couplings. How the signal scales with the number of spins is open, and reservoirs whose signals concentrate need exponentially many measurements [42].

Decay channels. We compare eight dissipators, sets of jump operators that we call decay channels, at equal total coupling $\mathrm{Tr}\, K$ with $K = \sum_k L_k^\dagger L_k$, and hold the Hamiltonian fixed to isolate the effect of the form of the dissipation. The Liouvillian, the generator of Eq. (2), has 4^N modes, patterns of change in the state that each decay at their own rate. Equal total coupling fixes their mean rate for every Hamiltonian, but not the slow modes, which hold a stored signal longest and become slower or faster with the channel (Appendix A.2, Table 2).

We discuss four of them (Fig. 1b). With independent decay every spin loses its excitation on its own. With collective decay all spins lose it through one shared jump operator, with equal phases or with opposite phases on spins 4 and 5, and we refer to the equal-phase and the opposite-phase channel. With dephasing every spin loses its phase. The channels differ in the kernel of the dissipator, the operators that the dissipation alone leaves unchanged, although the Hamiltonian can still turn them or couple them to decaying ones. For independent decay the kernel is the ground state alone. A collective channel cannot empty its dark states, so its kernel contains every operator on the dark subspace. There are $\binom{N}{\lfloor N/2 \rfloor}$ dark states, ten for five spins, and their number grows exponentially with the number of spins. Dephasing leaves the maximally mixed state unchanged whatever the input, so the driven reservoir relaxes to this state and stores nothing.

3 Results

3.1 Storing: the decay channel sets the memory lifetime

Increasing the loss separates the channels into two groups (Fig. 2). Without dark states the memory shrinks. With independent decay, recall retreats to shorter delays and the linear memory capacity falls steadily (Fig. 2a,c), and exchange and gain/loss behave in the same way (Fig. 2c, grey). With dark states the memory survives strong loss. With the equal-phase channel the capacity first dips and then doubles, and inputs fifty steps back are still recalled (Fig. 2b,c). The decay rates explain the split. Without dark states the Liouvillian gap, the slowest decay rate, grows in proportion to γ and the memory lifetime shrinks. With dark states the slow rates fall as $1/\gamma$ and the lifetime grows (Fig. 2d). Among the long-lived channels the capacity grows with the loss for the equal-phase channel and for collective decay in two groups, and returns to its weak-loss level for the opposite-phase channel and for pairwise decay (Fig. 2c, grey).

Zeno protection. Stronger collective decay prolongs the memory through the quantum Zeno effect [25, 43]. In the two-spin picture of Fig. 1c, the dark state $|-\rangle \propto |10\rangle - |01\rangle$ can decay only through the bright state $|+\rangle \propto |10\rangle + |01\rangle$, and strong loss suppresses this escape. Its rate falls as Ω^2/γ, where Ω couples $|+\rangle$ and $|-\rangle$, as the slow mode of an overdamped oscillator settles more slowly the stronger the friction.

For any number of spins the protection follows from perturbation theory in $1/\gamma$ at fixed system size, the adiabatic elimination of the bright states [26, 28]. At strong loss the reservoir follows its Hamiltonian restricted to the dark states, which turns the stored state without making it decay. Decay enters at the next order, when the Hamiltonian briefly lifts a dark state into a bright one. One step is then a rotation plus a loss that falls as $1/\gamma$ (Eq. (A2)), a nearly orthogonal map with a small decay rate, which is the classic recipe for a long memory in a network [33]. Calculations for three to six spins show the same decay out of the dark states with different prefactors, and with the equal-phase channel the capacity also grows with the loss at four and six spins (Appendix E.1).

Where the memory is stored. The dark states hold the memory, as two tests show (Appendix E.1). A model of the ten dark states alone, with no fitted parameter, reproduces the recall of the full reservoir at strong loss (Fig. 3a). Replacing the dark-state components of the state by their time averages after every step, which removes their dependence on the inputs, destroys all memory beyond a few steps, while the same operation on a randomly rotated set of equal size leaves most of it (Fig. 3b).

Added independent decay. Residual independent decay limits the lifetime. For the equal-phase channel, a relaxation time of T steps requires a

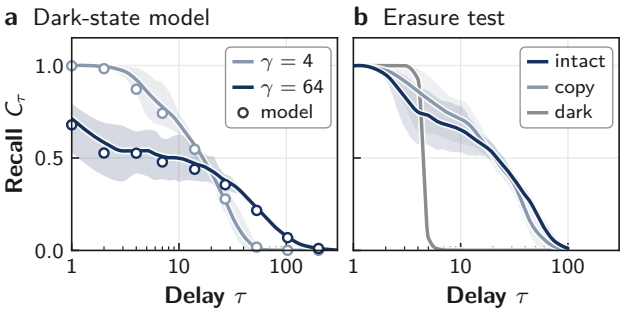


Figure 3: **Dark states hold the memory.** Equal-phase channel. **a** A model of the dark states alone (circles) reproduces the recall of the full reservoir (lines). **b** Erasing what the dark states know about the inputs destroys the memory (dark), while erasing a rotated copy of them leaves most of it (copy), at $\gamma = 16$.

ratio of collective to independent decay rates of at least $4.2\,T^2$ (Appendix E.6), so that a ratio of 10^4 protects 49 steps. For superconducting qubits coupled to a waveguide, a four-qubit dark state outlives the decay of a single qubit into the waveguide by more than two orders of magnitude [44], and qubit arrays reach a cooperativity of 172 [45].

Dark states protect a stored signal against decay but not against the inputs that follow it. With the opposite-phase channel the dark states are just as protected as with the equal-phase channel, yet the capacity does not grow.

3.2 Writing: new inputs mix the stored signal

Each new input turns the stored state by a small angle that depends on the input (Fig. 1d), so every newer input acts as noise on an older one. Over the memory lifetime the stored direction takes a random walk of small turns $\pm\theta$ and spreads by about $\theta\sqrt{t_{\mathrm{mem}}}$. Once the spread reaches about a radian, an old input survives only multiplied by the inputs that followed it, and a linear readout cannot turn such products back into single inputs. We call this the mixing of inputs. It is the nonlinearity in the trade-off between memory and nonlinearity [29, 30].

Counting the inputs. A stored signal splits into parts that each depend on a fixed set of inputs (Appendix A.4). The number of inputs in the set is the order D of the part. A function of u_{t-3} alone has $D = 1$, even if it is nonlinear in u_{t-3}, and the product $u_{t-3}u_{t-5}$ has $D = 2$. Linear memory requires $D = 1$. The total capacity divides among the orders [29, 46], and the share of single inputs is the part with $D = 1$.

The mixing law. Two quantities decide the order of a stored signal. The memory lifetime t_{mem} says how long the signal lasts. The mixing fraction s, between 0 and 1, is the part of the signal that one input turns differently from the average input (Eq. (B2)), and for small turns it grows as the square of the turn. In each step a fraction s of the stored signal picks up one more input, and a fraction $1/t_{\mathrm{mem}}$ of it is lost. A signal collects about $s\,t_{\mathrm{mem}}$ inputs before it is lost, and our first theorem makes this exact.

> **Theorem 1** (Mixing law, made precise in Appendix B)**.** Let a reservoir lose its stored signal slowly: in each step it is reset with a small probability and is otherwise rotated by the input, under conditions (1) to (3) of Appendix B. Pick a part of the stored signal with probability equal to its share of the total capacity. As the memory lifetime grows, its order D becomes exponentially distributed, with the mean
>
> $$\mathbb{E}D = s\,t_{\mathrm{mem}}$$
>
> up to an additive constant.

Scope of the law. The proof uses a reservoir with a reset, in any dimension, and nothing in it is quantum. Classical state-affine reservoirs [47] obey the law under the same conditions. For reservoirs without a reset the same proof applies, with the slow decay in place of the reset, if a single direction of the stored covariance is slow. Several slow directions keep distinct mixing scales (Appendix B.3).

Five spins. For five spins, Eq. (A3) gives the distribution of D exactly, and we compare it with the law (Fig. 4c). At a fixed amplitude, $\eta = 1$, the mixing fraction lies between 0.01 and 0.05 wherever the memory exceeds ten steps. As the memory grows, the mean of D follows $s\,t_{\mathrm{mem}}$ to within about two inputs, up to 23 inputs at the longest memory, and the distribution of D approaches the exponential law (Appendix B.3). The product $s\,t_{\mathrm{mem}}$ is the size of the trade-off between memory and nonlinearity [31], and dissipation enters it through both factors.

At a fixed amplitude a longer memory lowers

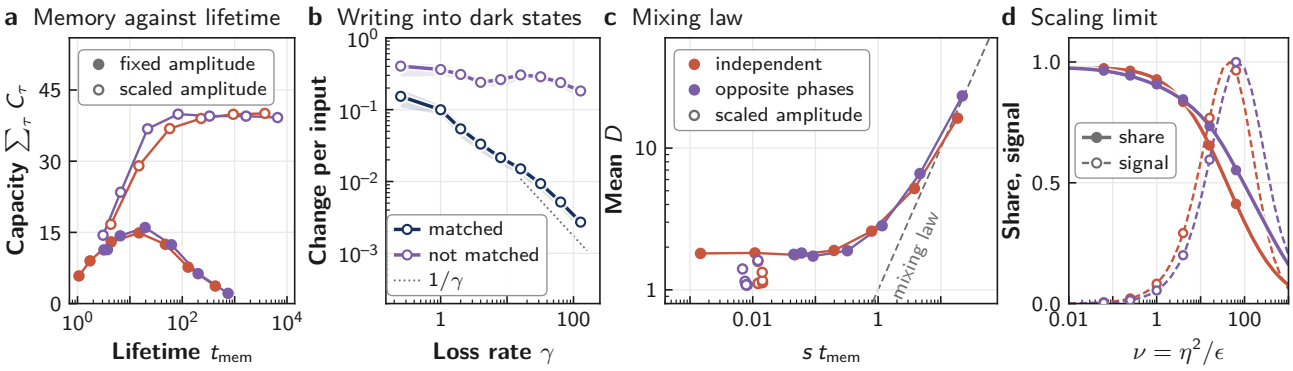


Figure 4: **New inputs mix the stored signal, and a scaled amplitude limits the mixing. a** At a fixed input amplitude the linear memory capacity peaks and then falls as the memory grows, and a scaled amplitude holds it at 40 units. **b** An input that is mode-matched to the equal-phase channel writes less into the dark states as the loss grows, while an input that is not mode-matched keeps writing. **c** The mean order follows the mixing law, and a scaled amplitude keeps it near one. **d** The scaling limit (lines) against the exact five-spin calculation (markers) for one reservoir, without a fitted parameter. Solid lines and filled markers show the share of single inputs in the total capacity, dashed lines and open markers the variance that single inputs leave in the features, scaled to its maximum. Panels a and c show exact moments, averaged over four reservoirs, two at the longest memories.

the linear memory capacity once $s\,t_{\mathrm{mem}}$ exceeds 0.2 to 0.3 (Fig. 4a). The capacity peaks at 16 units with the opposite-phase channel and at 15 units with independent decay, and falls to 2 and 4 units at the longest memories (Appendix E.3). The stored signal has moved into products of inputs, and with the opposite-phase channel the share of single inputs falls from 0.41 to 0.07. Single inputs stay readable only if the mixing fraction shrinks as the lifetime grows. The phases of the decay channel and the amplitude of the input both achieve this.

Channel phases. The phases of a collective channel decide whether an input reaches the dark states. Spin i couples to the channel with the sign $c_i = \pm 1$, and the collective lowering operator $\sum_i c_i \sigma_i^-$ annihilates the dark states. An input field is mode-matched to the channel if its strength on spin i is proportional to c_i. A mode-matched field $\sum_i c_i X_i$ is the lowering operator plus its adjoint and cannot couple two dark states, for any number of spins and any Hamiltonian. It writes into them only through brief excursions into bright states, with a strength that falls as $1/\gamma$. The part of a field that is not mode-matched writes into them directly (Fig. 4b, Appendix E.2).

The uniform input of Eq. (1) is mode-matched to the equal-phase channel. Its mixing fraction shrinks as the loss grows, and this channel reaches twice the capacity of the other two at a fixed amplitude (Table 3). With the opposite-phase channel the same input is not mode-matched. It writes directly into the dark states and mixes them at every loss rate. The two channels differ in nothing else. Rotating spins 4 and 5 by Z maps the opposite-phase channel with the uniform input onto the equal-phase channel with an input of opposite sign on these two spins, and changes only the signs of couplings J_{ij}, whose distribution is symmetric. The missing mode matching is the generic case, because a phase error opens a direct coupling within the dark subspace (Appendix E.2).

Input scaling. The input amplitude η controls the mixing for every channel. A weak input is known to favour linear memory [30, 31, 35], because it suppresses the products of the latest inputs that the reservoir forms within a few steps. The mixing law adds what a long memory requires. The mixing fraction grows as η^2, and the lifetime as the inverse of the leak rate ϵ at which the stored signal decays. The product $s\,t_{\mathrm{mem}}$ stays fixed only if the ratio $\nu = \eta^2/\epsilon$ does. We call an amplitude with fixed ν, which falls as $1/\sqrt{t_{\mathrm{mem}}}$, a scaled amplitude, and $\eta = 1$ the fixed amplitude. Theorem 1 shows that the scaling is necessary, because at a fixed amplitude the share of single inputs vanishes as the memory grows. Our second theorem shows that it is sufficient.

> **Theorem 2** (Scaling limit, made precise in Appendix B.4)**.** Let one step of a reservoir be a rotation, a leak at a small rate ϵ and an

input-dependent turn of amplitude η, under the decay condition of Appendix B.4. Let the leak vanish while ν stays fixed. Then the memory lifetime grows without limit, while the distribution of the order D converges. Its mean stays finite, and single inputs keep a fixed share of the total capacity. Both depend on the input only through ν. For large ν the mean of D grows in proportion to ν, as the mixing law requires.

The leak rate is $\epsilon = \gamma$ for independent decay and $\epsilon = 1/\gamma$ for the dark states, and the limit follows from the Hamiltonian and the jump operators without a fitted parameter. One qubit gives a closed form (Appendix C). For three spins with the opposite-phase channel, interval arithmetic confirms both theorems at strong loss. At a fixed amplitude single inputs lose their share of the capacity, and with a scaled amplitude they keep it. The stored signal resides there in coherences between the energy levels of the dark states (Appendix D).

Comparison with five spins. For five spins we use $\nu = 1$, an amplitude of $1/\sqrt{\gamma}$ for the opposite-phase channel and of $\sqrt{\gamma}$ for independent decay. The limit reproduces the exact five-spin calculation over three decades of ν, with the share of single inputs agreeing to within 0.01 (Fig. 4d, Appendix E.3). At $\nu = 1$ single inputs carry about 0.9 of the total capacity, and the linear memory capacity settles at 40 units for both channels as the lifetime grows (Fig. 4a). At the best loss rate of a fixed amplitude it is 15 units with independent decay and 16 with the opposite-phase channel. Simulated runs on eight reservoirs, with a readout trained on a finite number of inputs and summed over the first 300 delays, reach 35 units (Fig. 2c, Table 3).

The total capacity is fixed, so the gain moves capacity from combinations of inputs to single inputs. A task that needs products of inputs calls for a larger amplitude, and $s\,t_{\mathrm{mem}}$ gives the number of inputs that a stored signal then combines. A long memory needs no dark states, because weak independent decay with a scaled amplitude reaches one too. Dark states add a memory that stays long under strong dissipation and grows with the loss.

The cost of a weak input. The scaled amplitude is an upper bound on the input and not an optimum. With exact expectation values the capacity keeps rising as the input weakens, towards the number of features (Appendix E.4), because a weaker input mixes less and a fainter signal costs nothing without measurement noise. Theorem 2 quantifies what a weak input gives up (Fig. 4d). The variance that single inputs leave in the features vanishes in proportion to ν for weak inputs. It is largest where a stored signal combines a few inputs and single inputs carry about half of the total capacity (Appendix E.3).

3.3 Reading: the stationary state sets the measurable signal

A weaker input leaves a fainter signal. This costs nothing with exact expectation values and many measurements otherwise. Each feature is estimated from n shots per measurement setting and step, and its statistical error must stay below the signal, the variation of the feature with the inputs (Appendix A.5).

Signal size. The stationary state sets the size of the signal. An input acts on the stationary state ρ_*, the steady state of the dynamics averaged over the inputs, through the commutator with the input field, and only the part of this change that lies on the slow modes is remembered. Its size is the linear response of the stationary state to the input field (Appendix E.5), in the spirit of the link between memory and optical absorption of Ref. [9].

To compare all eight channels we choose the loss rate of each so that its relaxation time is 32 steps, and set $\eta = 0.35$. The relaxation time replaces the memory lifetime here because it does not depend on the input. The variance of the features then follows the linear response over four orders of magnitude for the seven channels that respond (Fig. 5b). The fraction of the exact capacity that 10^6 shots recover follows it as well (Fig. 5c). The rank correlation is 0.96 over all reservoirs and at least 0.81 within six of the seven channels (Appendix E.5). The equal-phase channel leaves a polarized state that the uniform input moves only through bright states. It has the largest exact capacity of all channels, and 10^6 shots recover less than 1% of it. Reversing the coupling phase of two spins lets the input move the state directly,

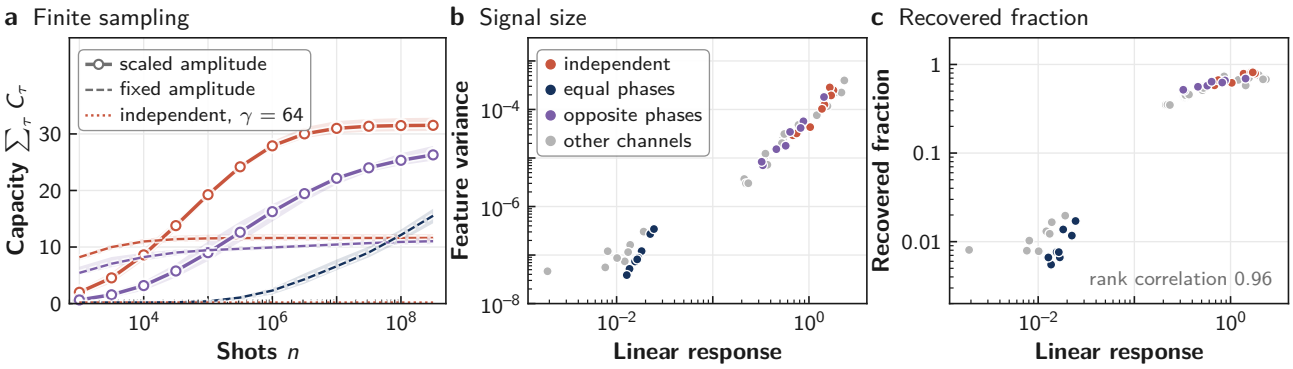


Figure 5: **Finite sampling limits the memory that can be read out. a** Capacity recovered against the number of shots per setting and step with the rewinding protocol, for independent decay at weak loss ($\gamma = 1/32$) and for the opposite-phase and the equal-phase channel at strong loss ($\gamma = 64$). **b** The variance of the features grows with the linear response of the stationary state to the input field. **c** So does the fraction of the exact capacity that 10^6 shots recover. Points in b and c are single reservoirs at $\eta = 0.35$. Their loss rates give every channel the same relaxation time, $t_{\mathrm{rel}} = 32$ steps, and every shot repeats the full input history.

and the opposite-phase channel recovers 60% of its exact capacity, against 73% for independent decay. The exact capacity does not predict what measurements recover. The linear response predicts which fraction of it they do.

Strong loss. The form of the dissipation matters most where the loss is strong (Fig. 5a). At $\gamma = 64$ and a fixed amplitude, 10^4 shots recover 8 units with the opposite-phase channel and nothing above the noise floor with the equal-phase channel or with independent decay (Table 3). A collective channel that is not mode-matched to the input is the right choice when strong loss is given, as for emitters that radiate into a common mode [22–24]. Where the loss can be made weak, independent decay recovers more, 11 units from the same number of shots. A short memory needs few shots, and the scaled amplitude pays off only with more. Its capacity overtakes that of the fixed amplitude between 10^4 and 10^5 shots (Appendix E.5).

Total cost. A measurement destroys the state, so every shot restarts the reservoir and repeats the recent inputs before the measurement, the rewinding protocol [10, 13]. The number of repeated steps grows with the memory lifetime. When every repeated step is counted (Appendix A.5), a budget of 10^9 reservoir steps per output step gives the three channels the same capacity at a fixed amplitude, each at its best loss rate (Table 3). The opposite-phase channel holds at a hundredfold stronger loss what independent decay holds at weak loss. With the scaled amplitude, weak independent decay recovers the most, 34 units against 21 with the opposite-phase channel.

With a finite number of shots a best amplitude exists. It falls as a power of the relaxation time, with fitted exponents between -0.40 and -0.46, close to the $-1/2$ of the scaled amplitude (Appendix E.4). The lifetime bounds the input amplitude from above, and the measurement budget bounds it from below.

4 Conclusion

Until now, the usual control over the dissipation of a quantum reservoir was its rate. Stronger loss meant a shorter memory, and the best rate was a compromise [6, 9]. Three facts were known separately. Dark states are protected by the quantum Zeno effect, memory trades against nonlinearity, and a weak signal is costly to measure. How they act together on a memory that keeps receiving inputs was open.

We now know that the form of the dissipation decides how such a memory is stored, written and read. Loss can protect a memory, because with collective decay a stronger loss makes it last longer. The trade-off between memory and nonlinearity has a law, which gives the number of inputs that mix into a stored signal as its lifetime times the fraction that one input mixes. An input scaled to the lifetime keeps this number fixed, so a memory can last as long as desired and still hold single inputs. Both statements are theorems, and they hold for classical and quantum reservoirs alike. What measurements recover of a memory

is decided by the stationary state.

For a reservoir under strong loss, three rules follow. Store the memory in dark states. Write it with an input that reaches these states and is scaled to the lifetime. Read it where the stationary state responds to the input.

Engineered dissipation is known to prepare and protect quantum states [18, 19]. Our results show that it also shapes the memory of a system that processes information over time, and that what this memory holds follows from the equations of motion, before any training. Dissipation makes a reservoir forget, and its form decides what the reservoir remembers.

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Data availability

The code, the data behind every figure, and the derivation and certificates of the three-spin limits are at https://github.com/eybmits/what-a-dissipative-quantum-system-remembers.

Author contributions

M.B. contributed to conception, methodology, software, analysis, visualization and writing, I.F. and J.W. to methodology, interpretation and editing, and C.L.-P. and J.S. to supervision and conceptual development. OpenAI Codex and Claude Code (Anthropic) assisted with manuscript preparation, checks of derivations and references, and simulation, verification and plotting code. The authors reviewed the outputs and take responsibility for the final content.

A Methods

This appendix specifies the five-spin simulations. Table 1 lists the time scales.

A.1 Reservoir, readout and capacity

The Hamiltonian of Eq. (1) uses $h = 1/2$ and couplings J_{ij} drawn uniformly from $[-1, 1]$, so the transverse field runs from h to $2h$. Each input u is drawn uniformly from $[0, 1]$ and held for a step of length $\Delta t = 1/2$. We integrate Eq. (2) in the 1024-dimensional basis of Pauli operators and evaluate the step map $E(u) = e^{\Delta t \mathcal{L}(u)}$ by Chebyshev interpolation in u at ten nodes. Each reservoir draws its couplings and its input stream from one seed, and every channel uses the same eight seeds. After a washout of $\max(800, 60\gamma)$ steps, at least four times the recall time, we record the 45 features, the expectation values of X_i, Y_i, Z_i and of X_iX_j, Y_iY_j, Z_iZ_j, for 2000 training, 400 validation and 1600 test inputs. The readout is a least-squares fit with an intercept on the training and validation inputs, and C_τ is the squared correlation of its estimate with $u_{t-\tau}$ on the test inputs. In these simulated runs the linear memory capacity sums C_τ over delays 1 to 300, and the exact moments of Appendix A.4 give the sum over all delays. A delay without any recall reads about $1/1600$ on the test inputs, so the sum has a floor of about 0.2 units. Unless a caption states otherwise, figures show the mean of the eight reservoirs, with bands from the 16th to the 84th percentile.

Table 1: Time scales, in input steps of length Δt.

t_{mem}	memory lifetime: decay time of the variance of the stored signal (Appendix A.4). It depends on the inputs and enters both theorems.
t_{rel}	relaxation time: decay time of the slow modes of the Liouvillian at the mean input (Appendix A.3). It does not depend on the inputs. Used in Secs. 3.1 and 3.3.
t_{f}	recall time: decay time of the response to one input, with $1/t_{\mathrm{f}} = 1/t_{\mathrm{rel}} + O(\eta^2)$ (Appendix A.3). Used in Appendix E only.
τ_{90}	rewinding window: the first τ_{90} delays hold 90% of the linear memory capacity (Appendix A.5)

A.2 Decay channels and equal coupling

Table 2 lists the jump operators of the eight decay channels. Their weights give every channel the same total coupling $\operatorname{Tr} K = N 2^{N-1}\gamma = 80\gamma$, with $K = \sum_k L_k^\dagger L_k$, which fixes the mean decay rate. For a traceless jump L the superoperator trace of its dissipator is $-2^N \operatorname{Tr} L^\dagger L$, while the commutator with any Hamiltonian is traceless, so the 4^N eigenvalues λ of the Liouvillian obey

$$\frac{1}{4^N} \sum \operatorname{Re} \lambda = -\frac{\operatorname{Tr} K}{2^N} \tag{A1}$$

for every channel and every Hamiltonian. The kernel of the dissipator is larger than the set of

Table 2: Decay channels at five spins, each with $\operatorname{Tr} K = 80\gamma$, and the number of operators that the dissipator leaves unchanged. The operators $\sigma_i^{\mp}$ lower and raise spin i, with $i < j$ for pairwise decay, $i \neq j$ for exchange and $c = (1, 1, 1, -1, -1)$.

channel	jump operators	kernel dimension
independent	$\sqrt{\gamma}\,\sigma_i^-$	1
equal phases	$\sqrt{\gamma}\sum_i \sigma_i^-$	100
opposite phases	$\sqrt{\gamma}\sum_i c_i\sigma_i^-$	100
two groups	$\sqrt{\gamma}\sum_{i\leq 3}\sigma_i^-$, $\sqrt{\gamma}\sum_{i>3}\sigma_i^-$	36
pairwise	$\sqrt{\gamma}\,\sigma_i^-\sigma_j^-$	36
exchange	$\sqrt{0.4\gamma}\,\sigma_i^-$, $\sqrt{0.3\gamma}\,\sigma_i^-\sigma_j^+$	1
gain/loss	$\sqrt{\gamma/1.3}\,\sigma_i^-$, $\sqrt{0.3\gamma/1.3}\,\sigma_i^+$	1
dephasing	$\sqrt{\gamma/2}\,Z_i$	32

dark states: the ten dark states of a collective channel carry $10 \times 10 = 100$ operators.

A.3 Slow modes, time scales and the effective Liouvillian

The input-averaged step $\bar{E} = \mathbb{E}_u E(u)$ is computed by six-point Gauss–Legendre quadrature. Its eigenvalues e_j give decay rates $\kappa_j = -\ln|e_j|/\Delta t$. If the kernel of the dissipator has dimension d_0, the slow modes are the $d_0 - 1$ slowest decaying ones besides the stationary state, and for $d_0 = 1$ the slowest mode. With their mean rate $\bar{\kappa}$ the recall time is $t_{\mathrm{f}} = 1/(\Delta t\,\bar{\kappa})$. The averaged step carries the part of a stored signal that still depends on a single input. Its mean rate adds decay and mixing to second order in the amplitude, $\bar{\kappa} = \bar{\kappa}_0 + c\,\eta^2 + O(\eta^4)$, because the inputs are distributed symmetrically about their mean and the term of $\bar{E}$ linear in η vanishes. Appendix E.2 tests this sum numerically. The relaxation time is the same quantity without the mixing, $t_{\mathrm{rel}} = 1/(\Delta t\,\bar{\kappa}_0)$, where $\bar{\kappa}_0$ is the mean decay rate of the slow modes of the Liouvillian at the mean input, $u = 1/2$. Figure 2d shows the decay rates $-\operatorname{Re}\lambda$ of this Liouvillian. With a collective channel we call the $d_0 - 1$ slowest of them dark and all faster ones bright.

Let $\mathcal{D}$ be the unit-rate dissipator, $\mathcal{P} = \lim_{t\to\infty} e^{t\mathcal{D}}$ the projector onto its kernel and $\mathcal{S} = (\mathcal{D} + \mathcal{P})^{-1} - \mathcal{P}$ its inverse on the decaying operators. At strong loss, adiabatic elimination of the decaying operators to second order [26, 28] gives the effective Liouvillian on the kernel,

$$\mathcal{G}(u) = \mathcal{P}\mathcal{H}(u)\mathcal{P} - \frac{1}{\gamma}\,\mathcal{P}\mathcal{H}(u)\mathcal{S}\mathcal{H}(u)\mathcal{P}, \qquad \text{(A2)}$$

with $\mathcal{H} = -i[H, \cdot\,]$. The first term is the Hamiltonian restricted to the kernel, which turns its operators, and the second is the loss through brief excursions into bright states.

For five spins the scaled amplitude has $\nu = 1$, which is $\eta = \min(1, \gamma^{-1/2})$ for the opposite-phase channel and $\eta = \min(1, \gamma^{1/2})$ for independent decay.

A.4 Number of inputs mixed

The order D of a part of the stored signal is the number of distinct inputs it depends on, the current input included (Sec. 3.2). Precisely, for independent inputs, every function of the past inputs with finite variance splits uniquely into orthogonal parts, each of which depends on a fixed finite set of inputs, the Hoeffding decomposition [48]. The order of a part is the size of its set, not a polynomial degree. For signs $u_t = \pm 1$ the parts are the products $\prod_{j\in J} u_{t-j}$, and for continuous inputs one uses orthogonal polynomials in each input. We collect the traceless coefficients of the state in an orthonormal operator basis in a vector r, on which quantum channels act affinely [49, 50], $r_t = A(u_t)r_{t-1} + b(u_t)$. All means and covariances are taken over input histories. Let $\bar{A} = \mathbb{E}A$ be the mean step, m the stationary mean and $\xi = A(u)m + b(u) - m$ the innovation, the change that one input writes, with covariance V. With $\mathcal{B}X = \bar{A}X\bar{A}^{\mathsf{T}}$, $\mathcal{T}X = \mathbb{E}[AXA^{\mathsf{T}}]$ and $\mathcal{K} = \mathcal{T} - \mathcal{B}$, the covariance G_k carried by products of order k obeys

$$G_1 = (1 - \mathcal{B})^{-1}V, \qquad G_{k+1} = (1 - \mathcal{B})^{-1}\mathcal{K}G_k. \qquad \text{(A3)}$$

The innovation adds the current input alone. The mean step carries a stored product along without adding an input. The fluctuating part $\mathcal{K}$ of the step multiplies it by the current input, which raises its order by one. The G_k add up to the stationary covariance Σ. A readout of features $y = Mr$ has the capacity $\operatorname{Tr}[(M\Sigma M^{\mathsf{T}})^{+}MG_kM^{\mathsf{T}}]$ of order k, and these capacities add up to the rank of the readout [29]. Divided by their sum, they give the share of each order, the distribution of D in Theorem 1. For one observable with coordinates a the share of order k is $a^{\mathsf{T}}G_ka/a^{\mathsf{T}}\Sigma a$. The share of single inputs is the share of order one, which includes the current input. Without the current input it is $a^{\mathsf{T}}\bar{A}G_1\bar{A}^{\mathsf{T}}a/a^{\mathsf{T}}\Sigma a$, the share of single past inputs. It is smaller by at most 0.08 in

the five-spin results and is the quantity bounded for three spins in Appendix D.

The memory lifetime and the mixing fraction come from the slowest mode of the covariance step $\mathcal{T}$: its eigenvalue is $1-1/t_{\mathrm{mem}}$, and with its right eigenvector Q and left eigenvector ℓ the mixing fraction is $s=\ell(\mathcal{K}Q)/\ell(Q)$, as in Appendix B.3. For five spins we evaluate Eq. (A3) on the 1023 traceless Pauli coordinates (Appendix E.3).

A.5 Finite sampling and the measurement budget

A quantum measurement gives a random outcome, so every feature is an average over n shots per measurement setting and step. We use three settings, with all spins read along X, Y or Z. For each step we form the outcome distributions of the three settings and draw n shots from each. The readout is fitted as above, with a ridge chosen from fourteen values between 0 and 10^4 on the validation inputs. In Sec. 3.3 the distributions come from the rewinding protocol [10]: for each output step the reservoir is restarted in $|00000\rangle$ and the last τ_{90} inputs are repeated, where the first τ_{90} delays hold 90% of the exact linear memory capacity. We count the budget B in reservoir steps, shots times repeated steps summed over the three settings, per output step, so that $n=B/(3\tau_{90})$, and we also run $n=10^3$ to $10^{8.5}$ in half decades. Figure 5b,c, Table 4 and the amplitude map of Appendix E.4 use the state with the full history instead. For the first two the loss rates are matched to a relaxation time of 32 steps (Appendix E.5).

B The mixing law and the scaling limit

This appendix proves the two theorems and extends the mixing law to reservoirs without a reset.

B.1 Setting and statement

We describe the reservoir by the d traceless coordinates r of its state in an orthonormal basis of operators, the Bloch vector for a qubit. A unitary acts on r as a rotation. In each step the reservoir is reset to a fixed state with a small probability p and is otherwise rotated by $O(u)$:

$$r_t=(1-p)\,O(u_t)\,r_{t-1}+p\,b. \tag{B1}$$

Here b represents the reset state. Equation (B1) averages over whether the reset occurred, so the only remaining randomness is the input history, with independent and identically distributed inputs u_t. A stored signal loses a fraction $1-(1-p)^2=2p-p^2$ of its variance per step, so its memory lifetime, defined by the slow eigenvalue $1-1/t_{\mathrm{mem}}$ of the covariance step (Appendix A.4), is $t_{\mathrm{mem}}=1/(2p-p^2)$, or $1/(2p)$ to leading order. Three conditions make the law hold:

(1) *The average step forgets.* The average rotation $A=\mathbb{E}O$ shrinks every vector: its spectral radius is below one.

(2) *The inputs mix.* The only covariance matrices that the random rotations leave unchanged on average, $\mathbb{E}[OXO^{\mathsf{T}}]=X$, are multiples of the identity.

(3) *The input writes.* One input changes the state: $\mathbb{E}\|(O-A)\mu\|^2>0$ with $\mu=(1-A)^{-1}b$.

The first condition means that only products of inputs can be stored for long, the second that a stored signal spreads evenly over all directions, and the third that there is something to store. The mixing fraction

$$s=1-\frac{\|A\|_F^2}{d} \tag{B2}$$

says how strongly one input turns the stored state. For a unit vector x of random orientation, $\mathbb{E}\|(O-A)x\|^2=s$: one input turns a fraction s of a stored signal away from the average rotation. The fraction lies between 0 and 1, the first condition makes it positive, and for small turns θ it grows as θ^2.

Mixing law, precise form. Assume conditions (1)–(3) and take any fixed linear readout. Pick a part of the stored signal with probability equal to its share of the readout's capacity, and let D be its order, the number of distinct inputs it depends on. As $p\to 0$, the ratio D/t_{mem} becomes exponentially distributed with mean s, and

$$\mathbb{E}D=s\,t_{\mathrm{mem}}+O(1).$$

Equivalently, products of more than $x\,t_{\mathrm{mem}}$ inputs carry the share $e^{-x/s}$ of the capacity.

B.2 Proof

The proof makes the competition of Sec. 3.2 exact: on average $1/s$ steps pass before the next input

joins a product, which meanwhile loses a fraction $1/(s\,t_{\mathrm{mem}})$ of its weight. The proof works with covariance matrices. Appendix A.4 splits the stationary covariance Σ of the state into the parts G_k carried by products of k inputs and gives their recursion, Eq. (A3): the mean step $\mathcal{B}$ carries a stored product along, and the fluctuating part $\mathcal{K}$ of the step adds one input to it. In the reset model the mean step is $\mathcal{B}X = (1-p)^2 AXA^{\mathsf{T}}$, the full step $\mathcal{T}X = (1-p)^2\,\mathbb{E}[OXO^{\mathsf{T}}]$, and $\mathcal{K} = \mathcal{T} - \mathcal{B}$. We write $\mathcal{B}_0$, $\mathcal{T}_0$ and $\mathcal{K}_0$ for the same maps at $p = 0$.

Step 1: the stored signal is small and spread evenly. The stationary mean is $m = p\,[1-(1-p)A]^{-1}b = p\mu + O(p^2)$, so one input writes the change $(1-p)(O-A)m$, of size p. Its covariance V has the trace $p^2v + O(p^3)$ with $v = \mathbb{E}\|(O-A)\mu\|^2$, which is positive by condition (3). The random rotations preserve the trace of a covariance and, by condition (2), leave only the identity unchanged. Since they cannot enlarge a covariance, the identity is a simple fixed point with the projector $X \mapsto (\operatorname{Tr} X/d)\,1$. On every other direction $(1-\mathcal{T})^{-1}$ stays bounded as $p \to 0$, while on the identity it equals $1/[1-(1-p)^2] \approx 1/(2p)$. The written covariance therefore piles up along the identity:

$$\Sigma = (1-\mathcal{T})^{-1}V = \frac{pv}{2d}\,1 + O(p^2). \qquad \text{(B3)}$$

Step 2: few inputs carry a vanishing share. The mean step forgets by condition (1), so $(1-\mathcal{B})^{-1}$ stays bounded as $p \to 0$. The recursion then makes $G_1 = (1-\mathcal{B})^{-1}V$ of order p^2, and the same holds for every G_k with fixed k. Measured against Σ, which is of order p and has full rank, any fixed set of orders carries a share of order p of the capacity of any readout. A long memory consists of products of many inputs.

Step 3: counting the inputs. Call $\mathcal{F} = (1-\mathcal{B})^{-1}\mathcal{K}$ the order transfer, so that $G_{k+1} = \mathcal{F}G_k$. At $p = 0$ it leaves the identity unchanged, because $\mathcal{K}_0 1 = 1 - AA^{\mathsf{T}} = (1-\mathcal{B}_0)1$. Its matching left eigenvector is the weight

$$\ell(X) = \operatorname{Tr}[(1-A^{\mathsf{T}}A)X], \qquad \ell(1) = ds, \quad \text{(B4)}$$

since $\ell(\mathcal{F}_0X) = \operatorname{Tr}\mathcal{K}_0X = \ell(X)$ when the rotations preserve the trace. For small p the eigenvalue one moves to

$$1 - \frac{2}{s}\,p + O(p^2)\,: \qquad \text{(B5)}$$

each step up in order costs a fraction $2p/s = 1/(s\,t_{\mathrm{mem}})$ of the weight, as in that competition. To see this, differentiate $(1-\mathcal{B})\mathcal{F} = \mathcal{K}$ at $p = 0$ and apply it to the identity, which gives $(1-\mathcal{B}_0)\mathcal{F}_0'1 = -2\cdot 1$. Its trace is the weight $\ell(\mathcal{F}_0'1) = -2d$, and dividing by $\ell(1) = ds$ gives the rate $2/s$.

To turn the eigenvalue into a distribution, weigh order k with $e^{-\zeta pk}$. Summing the recursion gives $\sum_k z^k G_k = z(1-z\mathcal{F})^{-1}G_1$, and as $z = e^{-\zeta p}$ approaches one, only the eigenvalue of Eq. (B5) makes the inverse large. With $\ell(G_1) = p^2v + O(p^3)$ and $\beta = 2/s$,

$$\sum_{k\ge 1} e^{-\zeta pk}G_k = \frac{pv}{2d}\,\frac{\beta}{\zeta+\beta}\,1 + O(p^2). \qquad \text{(B6)}$$

Divided by Eq. (B3), this is $\beta/(\zeta+\beta)$, the Laplace transform of an exponential distribution of pD with rate β. So $D/t_{\mathrm{mem}} = 2pD\,[1+O(p)]$ is exponential with mean s in the limit. Because both covariances are multiples of the identity to leading order, the same holds for the share of the capacity of any readout. Finally, $\sum_k kG_k = (1-\mathcal{F})^{-1}\Sigma = \Sigma/(\beta p) + O(p)$ gives $\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1)$.

Two technical points. On symmetric matrices, where covariances live, the eigenvalue one of $\mathcal{F}_0$ is simple: a fixed point satisfies $\mathcal{T}_0X = X$ and is therefore a multiple of the identity, and $\ell(1) > 0$ rules out a Jordan chain. For $p > 0$ every eigenvalue of $\mathcal{F}$ lies inside the unit circle, by a Perron–Frobenius argument: $\mathcal{F}$ maps positive matrices to positive matrices, and so does $(1-\mathcal{F})^{-1} = 1 + (1-\mathcal{T})^{-1}\mathcal{K}$. The eigenvalues of $\mathcal{F}_0$ therefore lie in the closed unit disk, only the simple one equals one, and only the eigenvalue of Eq. (B5) makes the sum diverge.

B.3 Reservoirs without a reset

A physical reservoir does not reset to a prescribed state, but the proof used the reset only through three facts: one covariance direction is slow, it leaks at a small rate, and the mean step forgets everything else. The same three steps therefore prove the law for general reservoirs.

Setting. One step is $r_t = A_u(\epsilon)r_{t-1} + b_u(\epsilon)$ on the coordinates that the inputs can reach, with a small leak ϵ, and $\mathcal{B}$, $\mathcal{T}$ and $\mathcal{K}$ are formed from A_u as before. We assume:

(i) A_u and b_u depend smoothly on ϵ in mean square, and b_u vanishes at $\epsilon = 0$.

(ii) The mean step forgets at $\epsilon = 0$: $\mathbb{E}A_u(0)$ has spectral radius below one.

(iii) The reservoir is mean-square stable for small $\epsilon > 0$: the covariance step $\mathcal{T}$ has spectral radius below one.

(iv) One covariance direction is slow: $\mathcal{T}$ has a simple eigenvalue $1-\kappa\epsilon+O(\epsilon^2)$ with $\kappa > 0$ and no other eigenvalue near one. At $\epsilon = 0$ its right eigenvector is a positive matrix Q, and its left eigenvector is a weight ℓ that is nonnegative on positive matrices, with $\ell(Q) = 1$.

(v) The input writes along it: the change written by one input, of order ϵ, has covariance $\epsilon^2 V_* + O(\epsilon^3)$ with $\ell(V_*) > 0$.

The reset model is the case $\epsilon = p$, $\kappa = 2$, $Q = 1$ and $\ell = \mathrm{Tr}\,/d$. Condition (iv) concerns the covariance that the random steps carry, not the number of slow modes: at a fixed amplitude the inputs turn the 99 slow dark modes of the opposite-phase channel differently in each step and spread a stored signal over all of them faster than it leaks, so that one covariance direction stays slow. Without (iv) one observable shows several order scales: two reset sectors with mixing fractions 1/4 and 3/4, selected by a classical flag, give products of more than $x\,t_{\mathrm{mem}}$ inputs the share $w\,e^{-4x}+(1-w)\,e^{-4x/3}$ with $w = 0.82$, which exact moments reproduce.

Mixing law without a reset. Assume (i)–(v) and take a fixed readout $y = Mr$ whose projected slow covariance MQM^{T} has full rank. Then the mixing fraction $s = \ell(\mathcal{K}_0 Q)$ is positive and at most one, and ϵD becomes exponentially distributed with rate κ/s. The mean order is $s/(\kappa\epsilon) + O(1)$: the mixing fraction times the memory lifetime $1/(\kappa\epsilon)$.

Proof. Steps 1 and 2 carry over. The stationary mean is of order ϵ by (i) and (ii), and the slow eigenvalue piles the written covariance up along Q,

$$\Sigma = \epsilon\,\frac{\ell(V_*)}{\kappa}\,Q + O(\epsilon^2),$$

and every fixed order carries a covariance of order ϵ^2. Here the readout condition enters: the shares of the orders in the capacity of the readout follow from Σ only if MQM^{T} has full rank. For one observable a this requires $a^{\mathsf{T}}Qa > 0$. In step 3, $\mathcal{F}_0$ has the same fixed points as $\mathcal{T}_0$, so Q spans them, and $\ell \circ (1-\mathcal{B}_0)$ is its left eigenvector, with value $\ell(\mathcal{K}_0 Q) = s$ on Q. Differentiating $(1-\mathcal{B})\mathcal{F} = \mathcal{K}$ at $\epsilon = 0$ and applying it to Q gives $(1-\mathcal{B}_0)\mathcal{F}_0' Q = \mathcal{T}_0' Q$, whose weight $\ell(\mathcal{T}_0' Q) = -\kappa$ is the rate of the slow eigenvalue of $\mathcal{T}$. The order eigenvalue is therefore $1-(\kappa/s)\epsilon + O(\epsilon^2)$, and the Laplace transform follows as before. It remains to show that s is positive, which also makes the eigenvalue simple. Because $\mathcal{K}_0$ maps positive matrices to positive matrices and ℓ is nonnegative on them, $\ell(\mathcal{B}_0 Y) \le \ell(\mathcal{T}_0 Y) = \ell(Y)$ for every positive Y, so $0 \le \ell(\mathcal{B}_0^j \mathcal{K}_0 Q) \le s \le 1$ for every j. If s vanished, the series $\mathcal{F}_0 = \sum_j \mathcal{B}_0^j \mathcal{K}_0$ would give $\ell(\mathcal{F}_0 Q) = 0$, which contradicts $\mathcal{F}_0 Q = Q$ and $\ell(Q) = 1$.

Numerical check. We compared the law with exact moments for 24 reset reservoirs and readouts in dimensions two to four. The largest deviation from the law shrinks in proportion to p, from 0.08 at $p = 3\%$ to 0.003 at $p = 0.1\%$. For five spins the share of the capacity in products of more than k inputs approaches $e^{-k/(s\,t_{\mathrm{mem}})}$. Its largest deviation, over k and the reservoirs, falls from 0.33 at $s\,t_{\mathrm{mem}} \approx 1$ to 0.07 at $s\,t_{\mathrm{mem}} \approx 20$ with the opposite-phase channel, and from 0.43 to 0.14 with independent decay.

B.4 The scaling limit

Setting. The amplitude now falls with the leak, $\eta^2 = \nu\epsilon$ at fixed ν, and uniformly in the input

$$\begin{aligned} A_u &= O_0\big[1 + \eta\,a_1(u) + \eta^2 a_2(u) - \epsilon Y\big] + o(\epsilon), \\ b_u &= \epsilon\,O_0 c_0 + o(\epsilon), \end{aligned} \tag{B7}$$

with a rotation O_0, $\mathbb{E}a_1 = 0$, and a step that is a rotation without the leak, so that $a_1^{\mathsf{T}} = -a_1$ and $a_2 + a_2^{\mathsf{T}} = a_1^2$. Weak independent decay has this form with $\epsilon = \gamma$, and the effective Liouvillian of Eq. (A2) has it on the dark coordinates with $\epsilon = 1/\gamma$: O_0 is the step of the Hamiltonian at the mean input, a_1 and a_2 are its response to the input field, and Y and c_0 the leak. Let Π_0 project onto the vectors that O_0 leaves fixed and Π onto the symmetric matrices that commute with O_0, the slow covariances. For q energy levels with distinct transition frequencies these are the covariances of the $q-1$ populations and one variance for each coherence, $q(q-1)$ numbers. On them, with

$\bar{a}_2 = \mathbb{E}a_2$,

$$\begin{aligned} \mathsf{L}X &= \Pi(YX + XY^{\mathsf{T}}), \\ \mathsf{S}X &= -\Pi(\bar{a}_2 X + X\bar{a}_2^{\mathsf{T}}), \\ \mathsf{K}X &= \Pi\,\mathbb{E}[a_1 X a_1^{\mathsf{T}}] \end{aligned} \tag{B8}$$

are the leak, what the inputs take from a stored product on average, and what they pass on to the next order, with $\operatorname{Tr}\mathsf{S}X = \operatorname{Tr}\mathsf{K}X$.

Scaling limit, precise form. Let every slow covariance decay: the eigenvalues of $\mathsf{L}+\nu(\mathsf{S}-\mathsf{K})$ have positive real parts. As $\epsilon \to 0$ at fixed ν:

(a) The stationary mean converges to the vector m_0 fixed by O_0 with $\Pi_0(Y - \nu\bar{a}_2)m_0 = \Pi_0 c_0$.

(b) With $V_1 = \mathbb{E}[a_1 m_0 m_0^{\mathsf{T}} a_1^{\mathsf{T}}]$ and $\mathsf{F}_\nu = \nu(\mathsf{L}+\nu\mathsf{S})^{-1}\mathsf{K}$,

$$\begin{aligned} \Sigma &\to \Sigma_\nu = \nu\,[\mathsf{L} + \nu(\mathsf{S}-\mathsf{K})]^{-1}\Pi V_1, \\ G_k &\to \mathsf{F}_\nu^{k-1}\,\nu\,(\mathsf{L}+\nu\mathsf{S})^{-1}\Pi V_1. \end{aligned}$$

(c) F_ν has a spectral radius $\varrho < 1$. For a readout $y = Mr$ with $M\Sigma_\nu M^{\mathsf{T}}$ of full rank, the distribution of D converges to that of (b), with a finite mean and the share of $k = 1$ for single inputs.

(d) As $\nu \to 0$, Σ_ν and ϱ vanish in proportion to ν. As $\nu \to \infty$, if the identity is the only slow covariance that the inputs leave unchanged, Σ_ν vanishes as $1/\nu$ and $\varrho/(1-\varrho) = \nu\,\mathbb{E}\|a_1\|_F^2/(2\operatorname{Tr}Y) + O(1)$, the $s\,t_{\mathrm{mem}}$ of the reset model.

Proof. Over a lifetime O_0 turns many times, so only what turns with it accumulates, and on that the leak and the inputs act at the same order ϵ. This is the weak-coupling limit of open quantum systems [51], taken for the covariance over input histories. It rests on a lemma: let R be orthogonal, P the projector onto the vectors it leaves fixed and $M_\epsilon = 1 - R - \epsilon RN + o(\epsilon)$. If $\Gamma = -PNP$ is invertible on the range of P, then $\epsilon M_\epsilon^{-1} \to \Gamma^{-1}P$. Indeed $PR = RP = P$, so in the blocks of P and $1-P$ the first diagonal block of M_ϵ is $\epsilon\Gamma + o(\epsilon)$, the off-diagonal blocks are of order ϵ, and the second tends to $1-R$, which is invertible there.

With $R = O_0$ the lemma gives (a), since $1-\bar{A} = 1 - O_0 - \epsilon O_0(\nu\bar{a}_2 - Y) + o(\epsilon)$. On symmetric matrices $\mathcal{R}_0 X = O_0 X O_0^{\mathsf{T}}$ is orthogonal, and Π projects onto its fixed points. With primes for the maps of Eq. (B8) without Π, Eq. (A3) has $\mathcal{B} = \mathcal{R}_0[1-\epsilon(\mathsf{L}'+\nu\mathsf{S}')]$, $\mathcal{K} = \epsilon\nu\,\mathcal{R}_0\mathsf{K}'$ and, because one input writes $\eta\,O_0 a_1 m_0$, $V = \epsilon\nu\,\mathcal{R}_0 V_1$, each up to $o(\epsilon)$. The lemma gives $\epsilon(1-\mathcal{B})^{-1} \to (\mathsf{L}+\nu\mathsf{S})^{-1}\Pi$ and $\epsilon(1-\mathcal{T})^{-1} \to [\mathsf{L}+\nu(\mathsf{S}-\mathsf{K})]^{-1}\Pi$, hence (b), and the order transfer $(1-\mathcal{B})^{-1}\mathcal{K}$ converges to a map whose nonzero eigenvalues are those of F_ν. The orders then fall geometrically at a common rate for small ϵ, which gives (c) once $\varrho < 1$.

The semigroups generated by $-(\mathsf{L}+\nu\mathsf{S})$ and $-[\mathsf{L}+\nu(\mathsf{S}-\mathsf{K})]$ keep positive matrices positive, as projections with the positive map Π of $X \mapsto e^{tW}Xe^{tW^{\mathsf{T}}}$, $W = \nu\bar{a}_2 - Y$, without and with the positive map $\nu\mathsf{K}'$ added to its generator, and the first is bounded by the second. The second decays by assumption, so both generators have positive inverses, and $(1-\mathsf{F}_\nu)^{-1} = 1 + \nu[\mathsf{L}+\nu(\mathsf{S}-\mathsf{K})]^{-1}\mathsf{K}$ is positive, which gives $\varrho < 1$ as in Appendix B.2. On covariances of fixed vectors $\mathsf{L}+\nu\mathsf{S}$ acts as $X \mapsto W_0 X + XW_0^{\mathsf{T}}$ with $W_0 = \Pi_0(Y - \nu\bar{a}_2)\Pi_0$, so W_0 is invertible and m_0 exists.

For (d), (b) is linear in ν to first order. For large ν, $\mathsf{F}_\nu \to \mathsf{S}^{-1}\mathsf{K}$, which leaves the identity unchanged because $\mathsf{S}1 = \mathsf{K}1$, with the left eigenvector $X \mapsto \operatorname{Tr}\mathsf{S}X$. To first order in $1/\nu$ its eigenvalue moves to $1 - 2\operatorname{Tr}Y/(\nu\,\mathbb{E}\|a_1\|_F^2)$. The mean m_0 falls as $1/\nu$, V_1 as $1/\nu^2$, and Σ_ν, which piles up along the identity as in Eq. (B3), as $1/\nu$. In the reset model $Y = 1$ and $s = \eta^2\,\mathbb{E}\|a_1\|_F^2/d$ to leading order.

C One qubit in closed form

A qubit relaxes towards $+X$ at a rate ϵ, through the jump $\sqrt{\epsilon}\,|+x\rangle\langle -x|$, is turned about Z by $u\theta$ per step by the inputs $u = \pm 1$, and is read along Y. The averaged step keeps the axes X and Y apart and the input-dependent part swaps them, so Y carries only odd orders: order $2j+1$ has the share $(1-\chi)\chi^j$ of its capacity, where χ is the factor by which the recursion (A3) returns covariance from Y through X to Y, and $\mathbb{E}D = (1+\chi)/(1-\chi)$. At a fixed angle with $\sin\theta \neq 0$, $1-\chi = 3\epsilon/\sin^2\theta + O(\epsilon^2)$ as $\epsilon \to 0$: ϵD becomes exponentially distributed with rate $3/(2\sin^2\theta)$, and the share of single inputs vanishes. If instead $\theta^2/\epsilon \to \nu$, Appendix B.4 applies with $O_0 = 1$ and

gives

$$\chi \to \frac{\nu^2}{(\nu+2)(\nu+1)}, \quad \operatorname{Var} Y \to \frac{4\nu}{(\nu+2)(3\nu+2)}. \tag{C1}$$

Single inputs keep the share $1-\chi$. The variance of Y is largest at $\nu = 2/\sqrt{3}$, where single inputs carry 80% of it, and vanishes as $4/(3\nu)$ for $\nu \gg 1$ and as ν for $\nu \ll 1$. Direct propagation of the density matrix reproduces these expressions.

D Three spins at strong loss

What is certified. Take three spins with a collective channel of opposite phases, Eq. (D1), four Hamiltonians H1–H4 with couplings drawn from fixed seeds, and every Hamiltonian whose three couplings lie in a small box around theirs. As γ grows:

- At a fixed input amplitude, the order of the memory of X_1, divided by γ, becomes exponentially distributed, and the variance of X_1 falls as $1/\gamma$ (boxes of half-width 1/32).
- With the scaled amplitude $\eta = 8/\sqrt{\gamma}$, that is $\nu = 64$, single past inputs (Appendix A.4) keep a finite share of the capacity of X_1, and the range of delays that holds 90% of their capacity grows in proportion to γ (boxes of half-width 1/512).

Model and check. The three spins evolve under

$$H_\eta(u) = \sum_{i<j} J_{ij} X_i X_j + \tfrac{1}{2} \sum_i Z_i + \tfrac{3+\eta u}{4} \sum_i X_i \tag{D1}$$

with one jump $\sqrt{\gamma}\,(\sigma_1^- - \sigma_2^- - \sigma_3^-)$, steps of $\Delta t = 1/2$ and signs $u = \pm 1$, at $\eta = 1$ or $\eta = 8/\sqrt{\gamma}$. The jump leaves three dark states, with six slow covariances. At strong loss one step of the full master equation is a rotation of them up to corrections of order $1/\gamma$, so the mixing law without a reset (Appendix B.3) gives the first statement and the scaling limit (Appendix B.4) the second, here with explicit error bounds for the full master equation. Both end in finitely many strict inequalities, which the archive `spin_certificates.zip` in the repository checks in interval arithmetic at 256 bits (ball arithmetic of Arb, python-flint 0.9.0), and the note `three_spin_limits.pdf` writes out the derivation. An independent program builds the full finite-rate maps and agrees with the certified values to better than one part in a million. The certified rate of the exponential law lies between 61 and 133, and with the scaled amplitude at $\gamma = 2^{16}$ single past inputs keep between 8% and 81% of the capacity of X_1.

Why coherence carries the memory. In the limit the mean m_0 consists of populations of the energy levels of the Hamiltonian restricted to the dark states, and a new input turns a population into a coherence between two levels: V_1 lies in the coherences. Coherence shapes the performance of spin reservoirs [52]. Dephasing in this energy basis after every step removes the coherences, and the stored signal falls to order ϵ^2, that is, as $1/\gamma^2$. Numerically the variance of X_1 stays between 10^{-4} and 2×10^{-3} from $\gamma = 2^8$ to 2^{18} when the coherences are kept and falls as γ^{-2}, by six orders of magnitude, when they are erased.

E Numerical details and supporting checks

E.1 Memory lifetime and dark states (Sec. 3.1)

Figures smooth recall curves for display only, by averaging each delay with its neighbours within a fifth of the delay, and no sum uses the smoothed values. The maps of Fig. 2a,b are computed at $\gamma = 0.1$, 0.25 and 2^{-1} to 2^6 and interpolated along the delay. Figure 2c stops the equal-phase channel and the two-group channel at $\gamma = 64$, where their recall at a fixed amplitude starts to reach past the 300 delays we sum. Summed over all delays with exact moments (Appendix A.4), their capacity keeps rising up to $\gamma = 256$.

Both tests of the dark states act on the 100 dark coordinates. For the dark-state model we write $\mathcal{P} = RR^\flat$ with $R^\flat R = 1$, where R embeds the dark coordinates α, evolve α under Eq. (A2) and record the same 45 features of the full state $(R - \mathcal{S}\mathcal{H}(u)R/\gamma)\,\alpha$. The readout is trained on them exactly as for the full reservoir, and the model has no fitted parameter (Fig. 3a). For the erasure test we replace the dark coordinates after every step by their time average, $r \mapsto r - RR^\flat(r - \bar{r})$. The control erases the dark subspace rotated by a random product of single-spin unitaries. Recall is measured at delays 1 to 100 with 800 training, 200 validation and 800 test inputs, at $\gamma = 16$ (Fig. 3b).

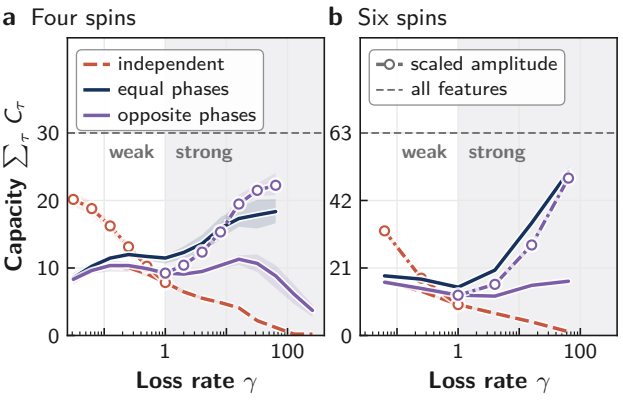


Figure 6: **Other reservoir sizes.** Capacity against the loss rate as in Fig. 2c, for four spins (eight reservoirs) and six spins (four reservoirs), with the same shading. A line stops where its recall reaches past the 300 delays summed.

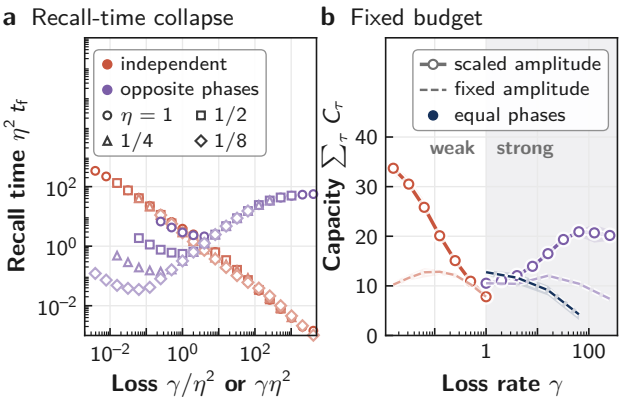


Figure 7: **Two supporting results. a** The recall time t_f, rescaled as the sum of decay and mixing predicts: the four amplitudes fall on one curve per channel. **b** Capacity recovered at a budget of 10^9 reservoir steps per output step, with every repeated step counted. Shading as in Fig. 2c.

The decay out of the dark states falls exactly as $1/\gamma$: for the equal-phase channel $\gamma\bar{\kappa}$ approaches $\kappa_c = 2.945$ from $\gamma = 512$ on, so t_f grows in proportion to γ, while it levels off near 56 steps with the opposite-phase channel and near 110 with pairwise decay. The second-order calculation of Eq. (A2) gives the same κ_c at five spins and 1.4, 2.1 and 3.4 at three, four and six spins: at every size we computed, the decay falls as $1/\gamma$, with a prefactor that depends on the size.

Figure 6 repeats the curves of Fig. 2c at four and six spins, with 30 and 63 features, on eight and four reservoirs. The opposite-phase channel reverses the sign of the last $\lfloor N/2 \rfloor$ spins. With the equal-phase channel the capacity grows with the loss at both sizes, from 11.5 to 18.4 units between $\gamma = 1$ and 64 at four spins and from 15 to 51 at six, and a scaled amplitude raises the best capacity of the other two channels. At three spins, which have only three dark states, the capacity of the equal-phase channel does not grow.

E.2 Recall time and phases (Sec. 3.2)

The recall time t_f shows the balance of decay and mixing. Its rate adds the decay of the stored signal and a mixing rate that grows as η^2 (Appendix A.3). The decay grows as γ with independent decay and falls as $1/\gamma$ with the opposite-phase channel, so $\eta^2 t_f$ depends on loss and input only through γ/η^2 and through $\gamma\eta^2$, the $1/\nu$ and ν of Appendix B.4. Both collapses hold for the four amplitudes $\eta = 1$, $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$ (Fig. 7a), for γ from 2^{-8} to 2^6 with independent decay and from 2^{-2} to 2^{12} with the opposite-phase channel.

The change per input of Fig. 4b is $2\|R^b[E(\frac{3}{4}) - E(\frac{1}{4})]r_*\|$ at the stationary state r_* of the equal-phase channel, for the input patterns $(1, \ldots, 1)$ and $\sqrt{2}\cos(2\pi i/5)$ multiplying X_i. The mode-matched pattern falls as $1/\gamma$, to 0.003 at $\gamma = 128$, while the other falls from 0.40 to 0.18. The rule is sharp: a phase error δ couples dark states directly at order δ, so beyond $\gamma \sim 1/\delta$ the equal-phase channel behaves like the opposite-phase channel.

E.3 Orders and capacities (Sec. 3.2)

The distribution of the number of inputs mixed, Eq. (A3), uses six-point Gauss–Legendre quadrature for the expectations over the input and solves the equations $X - \bar{A}X\bar{A}^\mathsf{T} = C$ in the eigenbasis of $\bar{A}$. The shares of the 45 features are summed until the rest falls below 10^{-8}, or until successive orders shrink at a settled ratio, whose geometric tail we add. The recall follows from the same moments, $C_\tau = c_\tau^\mathsf{T}\Sigma_y^{-1}c_\tau/\mathrm{Var}\,u$ with c_τ the covariance of the features with $u_{t-\tau}$ and Σ_y their covariance, and reproduces the simulated capacities to a median of 2%. Three steps of inverse iteration on $\mathcal{T}$, from the input's covariance V and from the identity, give t_{mem}, Q and ℓ. Loss rates run in powers of four over the ranges of Fig. 4c, and with the scaled amplitude to 16384 and 1/4096, on four reservoirs, two at the longest memory lifetimes. At a fixed amplitude the mean order of the opposite-phase channel rises from 1.8 at $\gamma = 1$ to 23 at $\gamma = 4096$, and with independent decay it

stays within 2.2 of $s\,t_{\mathrm{mem}}$ up to $s\,t_{\mathrm{mem}} = 18$.

Figure 4a shows the linear memory capacity over all delays, from the same moments. At a fixed amplitude it peaks at 16.0 units with the opposite-phase channel, at $\gamma = 64$ and $s\,t_{\mathrm{mem}} = 0.32$, and falls to 2.2 units at a lifetime of 750 steps, where single inputs carry 0.07 of the total capacity. With independent decay it peaks at 14.9 units, at $\gamma = 1/16$ and $s\,t_{\mathrm{mem}} = 0.20$, and falls to 3.7 units at 430 steps, where single inputs carry 0.10. With the scaled amplitude single inputs carry 0.85 of the total capacity at a lifetime of 20 steps and 0.88 to 0.92 beyond 80 steps, where the capacity stays between 39 and 40 units and $s\,t_{\mathrm{mem}}$ between 0.007 and 0.014.

The scaling limit. We build the limit of Appendix B.4 from the Liouvillian of each reservoir: O_0, a_1 and a_2 are the orders zero to two of the step in the input field, and Y and c_0 its first order in the leak. Independent decay has 992 slow covariances and the opposite-phase channel 90, for the ten dark states under Eq. (A2), and at $\nu = 1$ they decay by at least $0.5\,\epsilon$ and $1.4\,\epsilon$ per step. With the opposite-phase channel 33 directions of the 45 features respond at leading order, 11 through the admixture of decaying operators of order $1/\gamma$, which the readout of the limit includes and which raises the share of single inputs by up to 0.02, and one at order $1/\gamma^2$, which the recursion no longer resolves from $\gamma = 1024$ on.

For the effective Liouvillian of Eq. (A2) the recursion converges onto the limit: at $\nu = 1$ the share of single inputs agrees to four digits from $\gamma = 2^{14}$ on. For the full master equation at $\gamma = 1/4096$ and 4096, on two reservoirs and for $\nu = 1/16$ to 64, it differs from the limit by at most 0.002 with independent decay and 0.009 with the opposite-phase channel, the mean of D by 0.003 and 0.016, and the variance that single inputs leave in the 45 features by 0.4%. At $\nu = 1$ the limit gives single inputs 0.86 to 0.93 of the total capacity over four reservoirs with independent decay and 0.90 to 0.94 with the opposite-phase channel. The variance that single inputs leave is largest at $\nu = 22$ to 46 with independent decay and at $\nu = 68$ to 215 with the opposite-phase channel. There single inputs carry 0.38 to 0.55 of the total capacity, and $\varrho/(1-\varrho)$ lies between 0.8 and 4.8.

Table 3 collects the capacities of simulated runs on eight reservoirs. Their readout is trained and tested on a finite number of inputs, and their sum stops at 300 delays. On the four reservoirs that both use, with the scaled amplitude at $\gamma = 1/64$ and 256, the simulated runs give 35.0 and 37.1 units, and the exact moments give 36.8 and 39.4 units within the first 300 delays and 36.9 and 39.9 over all delays. All capacities of a readout add up to its rank, 45 here [29], so raising the linear memory capacity from 15 to 40 units leaves at most 5 instead of 30 for products of inputs.

Table 3: Linear memory capacity $\sum_\tau C_\tau$ of the five-spin reservoir in simulated runs, in units (maximum 45) and summed over the first 300 delays: with exact expectation values, at a budget of 10^9 reservoir steps per output step, and with n shots per setting and step with the rewinding protocol (Sec. 3.3). Mean and standard deviation over eight reservoirs. Exact moments over all delays, as quoted in Sec. 3.2, are higher by up to 4 units (Appendix E.3).

	independent decay	opposite phases	equal phases
fixed amplitude[a]	14.5 ± 1.0	14.7 ± 1.1	30.6 ± 1.4
scaled amplitude[b]	34.9 ± 2.0	35.7 ± 3.0	–[c]
budget 10^9, fixed[d]	12.8 ± 0.7	12.0 ± 0.7	12.8 ± 0.7
budget 10^9, scaled[e]	33.7 ± 1.1	20.9 ± 1.1	–[c]
$\gamma = 64$, exact[f]	1.2 ± 0.0	14.7 ± 1.1	30.6 ± 1.4
$\gamma = 64$, $n = 10^4$[f]	0.2 ± 0.0	8.2 ± 0.8	0.2 ± 0.0
$\gamma = 64$, $n = 10^6$[f]	0.2 ± 0.0	9.9 ± 0.7	2.3 ± 1.0

[a] At the loss rates with the largest sum over 300 delays, $\gamma = 0.1$, 64 and 64. Over all delays the capacity of the equal-phase channel keeps rising up to $\gamma = 256$ (Appendix E.1). [b] At $\gamma = 1/64$ and 256. [c] The uniform input is mode-matched to this channel. [d] At $\gamma = 1/8$, 16 and 1, the best loss rates on the grid. [e] At $\gamma = 1/64$ and 64. [f] At a fixed amplitude.

E.4 Best input amplitude (Secs. 3.2 and 3.3)

We mapped the linear memory capacity over loss rates and amplitudes: the first four reservoirs, independent decay at $\gamma = 2^{-8}$ to 2^{-2} and at 1, the opposite-phase channel at $\gamma = 1$, 4 and 2^3 to 2^8, and fifteen amplitudes from $\eta = 2^{-7}$ to 1, with exact features or 10^6 to 10^8 shots per setting and step from the state with the full history. With exact expectation values the capacity rises as the input weakens, up to 43 of the 45 units. With 10^6 to 10^8 shots an optimum exists. Its amplitude falls with the relaxation time, computed as the recall time at $\eta = 2^{-8}$, with fitted exponents between -0.40 and -0.46 in all six fits over the loss rates whose relaxation time is at least four steps. Without the loss rates whose recall reaches past the 300 delays summed, the exponents are

-0.48 to -0.56 with independent decay and -0.38 to -0.39 with the opposite-phase channel. The optimum is flat. Over the loss rates of the map, one amplitude recovers on average 91% to 97% of the capacity at the optimum tracked at each loss rate, so within such a range the amplitude needs no fine tuning.

E.5 Measurement (Sec. 3.3)

With the opposite-phase channel, the scaled amplitude lowers the variance of the features from 8×10^{-5} to 5×10^{-6}, and restarting each run from $|00000\rangle$, as the rewinding protocol does, raises it to 4×10^{-5}, against 1.2×10^{-4} with independent decay and the scaled amplitude at $\gamma = 1/32$.

The rewinding window τ_{90} grows with the memory lifetime: with the scaled amplitude of the opposite-phase channel from about 30 steps at $\gamma = 16$ to 130 at $\gamma = 256$. The thresholds of Sec. 3.3 are the crossings of the capacities at a fixed and with the scaled amplitude, from runs with the rewinding protocol at 10^3 to $10^{8.5}$ shots per setting and step. With the rewinding protocol the equal-phase channel at $\gamma = 64$ stays at the floor up to 10^4 shots and recovers 2.3 units at 10^6 and 15.5 at 3×10^8. There it passes the opposite-phase channel at a fixed amplitude, 11.0 units, and stays below it with the scaled amplitude, 26.3 units. At 10^3 and 10^4 shots the opposite-phase channel at $\gamma = 64$ recovers 5.4 and 8.2 units at a fixed amplitude, and independent decay at $\gamma = 1/32$ recovers 8.2 and 11.0. The scaled amplitude overtakes the fixed one at 2×10^4 shots with independent decay and at 10^5 with the opposite-phase channel, and independent decay then reaches 31.5 units at 3×10^8.

Figure 7b shows the capacity at a budget of 10^9 reservoir steps per output step, which is 3×10^6 to 3×10^7 shots per setting and step, and Table 3 lists its best values. With the full history, eight times more training inputs change the sampled capacity by less than 1.3 units.

For the matched relaxation time of Fig. 5b,c and Table 4 we find, for every channel and reservoir, the loss rate at which $t_{\mathrm{rel}} = 32$ steps, at weak loss when the kernel of the dissipator is the stationary state alone and at strong loss otherwise, and run the reservoir there with $\eta = \sqrt{4/32}$. The linear response of the stationary state is the Euclidean norm of the Pauli coefficients $\mathrm{Tr}(PW)$, over all Pauli strings P, of the change W: the commutator $-i[H_{\mathrm{in}}, \rho_*]$ with $H_{\mathrm{in}} = h\sum_i X_i$ and ρ_* the stationary state, projected spectrally onto the slow modes. These are the modes of the kernel, or without one those decaying at most four times faster than the slowest. The loss rate is matched for each reservoir, so Table 4 gives its mean. Over the 56 reservoirs of the seven channels that respond, the rank correlation of the linear response is 0.96 with the recovered fraction of the capacity and 0.83 with the recovered capacity itself. The seven channel medians have a rank correlation of 0.93 with the recovered fraction. Within single channels, over eight reservoirs each, it is 0.81 to 0.95 for six channels and 0.29 for pairwise decay, and with the variance of the features it is at least 0.83 for all seven.

Table 4: The eight decay channels at a relaxation time of 32 input steps and $\eta = 0.35$: loss rate, linear response of the stationary state, and linear memory capacity with exact expectation values and with 10^6 shots per measurement setting and step. Mean and standard deviation over eight reservoirs.

channel	loss rate γ	response	capacity exact	capacity at 10^6
independent	0.092	1.3	23.7 ± 1.1	17.4 ± 2.3
exchange	0.064	0.30	25.1 ± 0.8	10.1 ± 0.9
gain/loss	0.079	0.46	24.4 ± 0.8	12.5 ± 0.9
pairwise	168	1.5	18.2 ± 2.0	12.4 ± 1.2
opposite phases	101	0.68	25.3 ± 2.6	15.0 ± 1.5
two groups	73	0.011	26.6 ± 2.9	0.3 ± 0.1
equal phases	47	0.018	35.3 ± 0.8	0.3 ± 0.1
dephasing	143	0	0.0	0.2[a]

[a] A readout without any recall reads about 0.2 units, the floor of the sum over 300 delays.

E.6 Added independent decay and dephasing

Adding independent decay $\sqrt{\gamma_{\mathrm{ind}}}\,\sigma_i^-$ to the equal-phase channel gives the dark modes the mean decay rate $\bar\kappa_0 = \kappa_{\mathrm{c}}/\gamma + \kappa_{\mathrm{l}}\gamma_{\mathrm{ind}}$, where to first order κ_{l} is the trace of the added dissipator on the 99 slow dark modes divided by 99, $140/99 \approx 1.414$. Dephasing $\sqrt{\gamma_\phi}\,Z_i$ adds $\kappa_\phi\gamma_\phi$ with $\kappa_\phi = 403.2/99 \approx 4.073$, and the full spectra reproduce both coefficients to within 0.2%. A relaxation time of at least T steps, $\bar\kappa_0 \le 1/(\Delta t\, T)$, then needs $\gamma/\gamma_{\mathrm{ind}} \ge 4\kappa_{\mathrm{c}}\kappa_{\mathrm{l}}\,\Delta t^2 T^2 \approx 4.2\,T^2$, or $12\,T^2$ for dephasing.

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